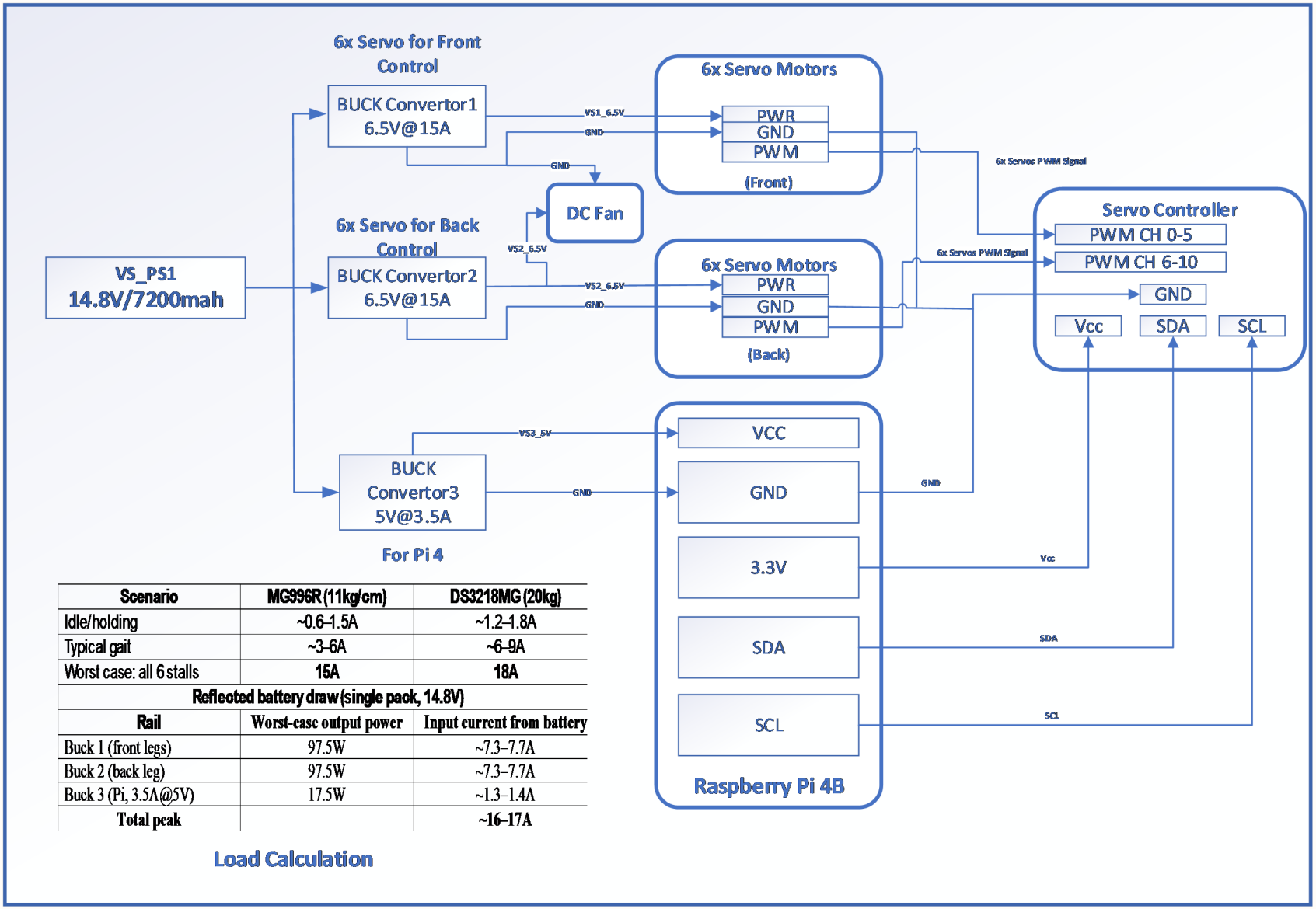
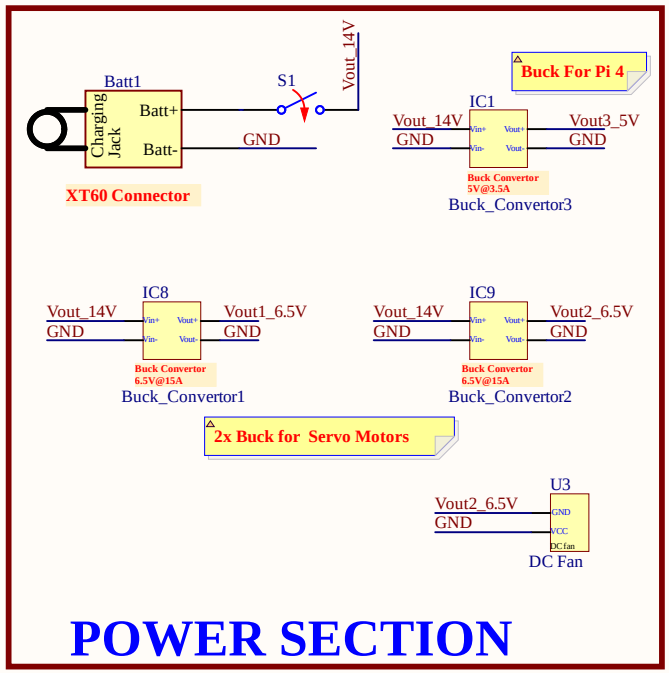
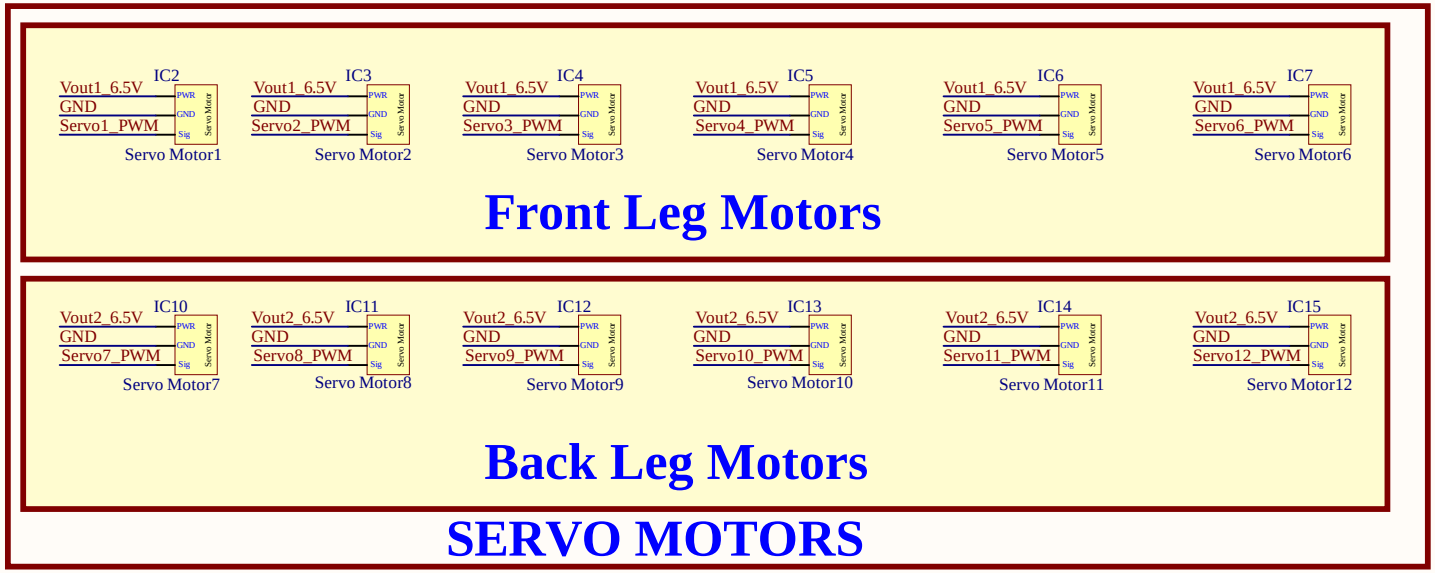


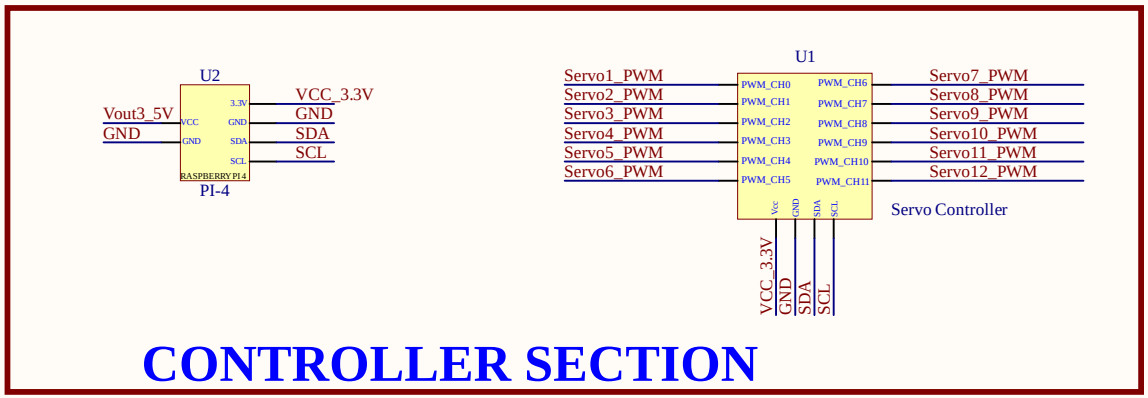
Scenario	MG996R (11kg/cm)	DS3218MG (20kg)
Idle/holding	~0.6–1.5A	~1.2–1.8A
Typical gait	~3–6A	~6–9A
Worst case: all 6 stalls	15A	18A
Reflected battery draw (single pack, 14.8V)		
Rail	**Worst-case output power**	**Input current from battery**
Buck 1 (front legs)	97.5W	~7.3–7.7A
Buck 2 (back leg)	97.5W	~7.3–7.7A
Buck 3 (Pi, 3.5A@5V)	17.5W	~1.3–1.4A
Total peak		**~16–17A**

Load Calculation

Title *Block Diagram*			**Designed by:** *M. Huzaifa Rahim Malik*
Size: A4	Number: 1	Revision: 1.0	
Date: 28/10/2026 Time: 8:45:27 am	Sheet 1 of 2		
File: D:\Github\12-DOF-Quadrupled-Dog\hardware\altium\Block Diagram.SchDoc			







Title *Quadrupled Design*			**Designed by:** *M. Huzaifa Rahim Malik*
Size: A4	Number:2	Revision:1.0	
Date: 28/10/2026	Time: 8:45:28 am	Sheet 2 of 2	
File: D:\Github\12-DOF-Quadrupled-Dog\hardware\altium\Schematic.SchDoc			